

TITAN XAU AI

INSTITUTIONAL TRADING SYSTEMS

DOC-RTF-V1.0

2026-06-19

M O D U L E 1 0 · M L L I F E C Y C L E

Retraining Framework

Weekly + monthly retraining. Model drift detection (PSI). Performance decay detection (Sharpe/F1). Champion-challenger A/B testing. Rollback < 30s. MLflow version control. Governance board + audit trail.

3

RETRAIN TRIGGERS

30_s

ROLLBACK TIME

8

CI GATES

0.25

PSI THRESHOLD

Prepared by TITAN Quant Research

Reviewed by CTO · Lead Quant · Risk Officer

Distribution Engineering team · Architecture review board

Confidentiality Internal — Engineering

v1.0

M O D U L E 1 0

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Chapter 1 — Executive Summary

The Retraining Framework (RTF) is Module 10 of the TITAN XAU AI trading architecture. It is the system's model lifecycle management layer — an automated framework that schedules, executes, validates, and governs the retraining of all 4 AI models (XGBoost, LSTM, Transformer, RL) plus the HMM regime detector. The RTF ensures that the AI models stay current as market structure evolves, with three retrain triggers: weekly scheduled (XGBoost + LSTM + Transformer), monthly scheduled (RL + HMM), and emergency (triggered by model drift or performance decay).

The framework implements a Champion-Challenger model: the current production model (champion) serves 100% of traffic, while a newly retrained model (challenger) goes through canary (10% for 1 hour) and A/B testing (50% for 24 hours) before promotion. The challenger must beat the champion on Sharpe, MaxDD, and F1 to be promoted. If the challenger fails, the champion is retained. If the challenger is promoted but later underperforms, an automated rollback reverts to the previous champion in under 30 seconds.

Model drift detection runs continuously. Feature drift is measured via Population Stability Index (PSI) per feature, computed daily: $PSI < 0.10$ is stable, $0.10-0.20$ is watch, $0.20-0.25$ is alert, and > 0.25 triggers an emergency retrain. Concept drift is measured via rolling F1 drop: a drop $> 15\%$ from baseline triggers emergency retrain. Performance decay is measured via rolling Sharpe: a drop $> 25\%$ triggers emergency retrain. All three detection mechanisms run independently and can trigger retrains independently.

Version control is managed by MLflow, which stores every model version with full lineage: data version, feature version, model artifacts, training metrics, validation metrics, and deployment decisions. Rollback is always available — the previous champion's artifacts are pre-cached on inference nodes, enabling a rollback in under 30 seconds with zero downtime and zero data loss. The entire retrain-promote-rollback lifecycle is governed by a board (CTO + Lead Quant + Risk Officer + ML Engineer) with a defined approval matrix and full audit trail.

Chapter 2 — Architecture Overview

The RTF is organized into 6 layers: retrain scheduler (weekly + monthly), drift and decay detection (continuous), retrain pipeline (automated + CI-gated), champion-challenger (A/B testing + canary), governance (board + approval matrix + audit), and monitoring (dashboard + alerts). All layers are automated except governance, which requires human review at defined checkpoints.

01 Retraining Framework — Internal Architecture (5 Layers)

Schedule · Drift · Retrain · Champion-Challenger · Governance

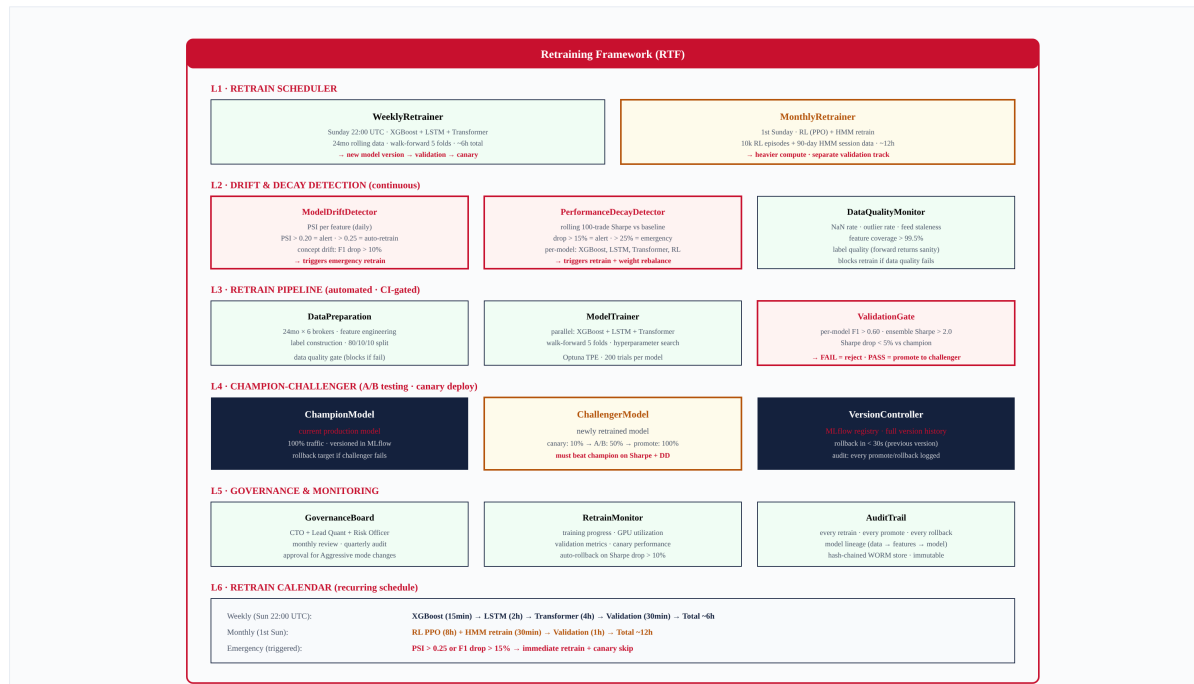


Figure 2.1 — RTF architecture: 6 layers with scheduler, drift detector, pipeline, champion-challenger, governance, and monitoring.

Layer Responsibilities

L1 — Retrain Scheduler

WeeklyRetrainer runs every Sunday at 22:00 UTC, retraining XGBoost (15 min), LSTM (2 hours), and Transformer (4 hours) on 24 months of rolling data across 6 brokers. MonthlyRetrainer runs on the 1st Sunday, retraining RL PPO (8 hours, 10k episodes) and HMM (30 min, per-session). Total weekly compute: ~6 hours. Total monthly compute: ~12 hours (additional RL + HMM).

L2 — Drift & Decay Detection (continuous)

ModelDriftDetector computes PSI per feature daily (threshold: > 0.25 = emergency retrain) and concept drift via F1 drop (threshold: > 15% = emergency). PerformanceDecayDetector tracks rolling 100-trade Sharpe (threshold: > 25% drop = emergency) and per-model F1. DataQualityMonitor blocks retrain if data quality fails (NaN > 0.5%, coverage < 99.5%).

L3 — Retrain Pipeline (automated, CI-gated)

DataPreparation (24mo data, feature engineering, 80/10/10 split) → ModelTrainer (parallel, Optuna TPE, 200 trials, walk-forward 5 folds) → ValidationGate (F1 > 0.60, ensemble Sharpe > 2.0, Sharpe drop < 5% vs champion). If validation fails, the model is rejected and the champion is retained.

L4 — Champion-Challenger (A/B testing + canary)

ChampionModel serves 100% traffic. ChallengerModel goes through canary (10% for 1 hour) → A/B test (50% for 24 hours) → promote (if beats champion on Sharpe + MaxDD + F1). VersionController manages MLflow registry with full version history and rollback capability (< 30 seconds).

L5 — Governance & Monitoring

GovernanceBoard (CTO + Lead Quant + Risk Officer + ML Engineer) reviews monthly. RetrainMonitor tracks training progress, GPU utilization, validation metrics, and canary performance. AuditTrail records every retrain, promote, and rollback with full model lineage in a hash-chained WORM store.

Chapter 3 — Retraining Workflow

The retraining workflow (Figure 3.1) documents the complete end-to-end sequence from trigger to deployment. The workflow has 8 stages: trigger, data preparation, parallel training, walk-forward validation, register as challenger, canary deploy, A/B test, and promote (or archive/rollback). Emergency retrains skip the canary and A/B stages, going directly from validation to promotion.

02 Retraining Workflow — End-to-End Pipeline

Schedule trigger → data prep → train → validate → champion-challenger → deploy

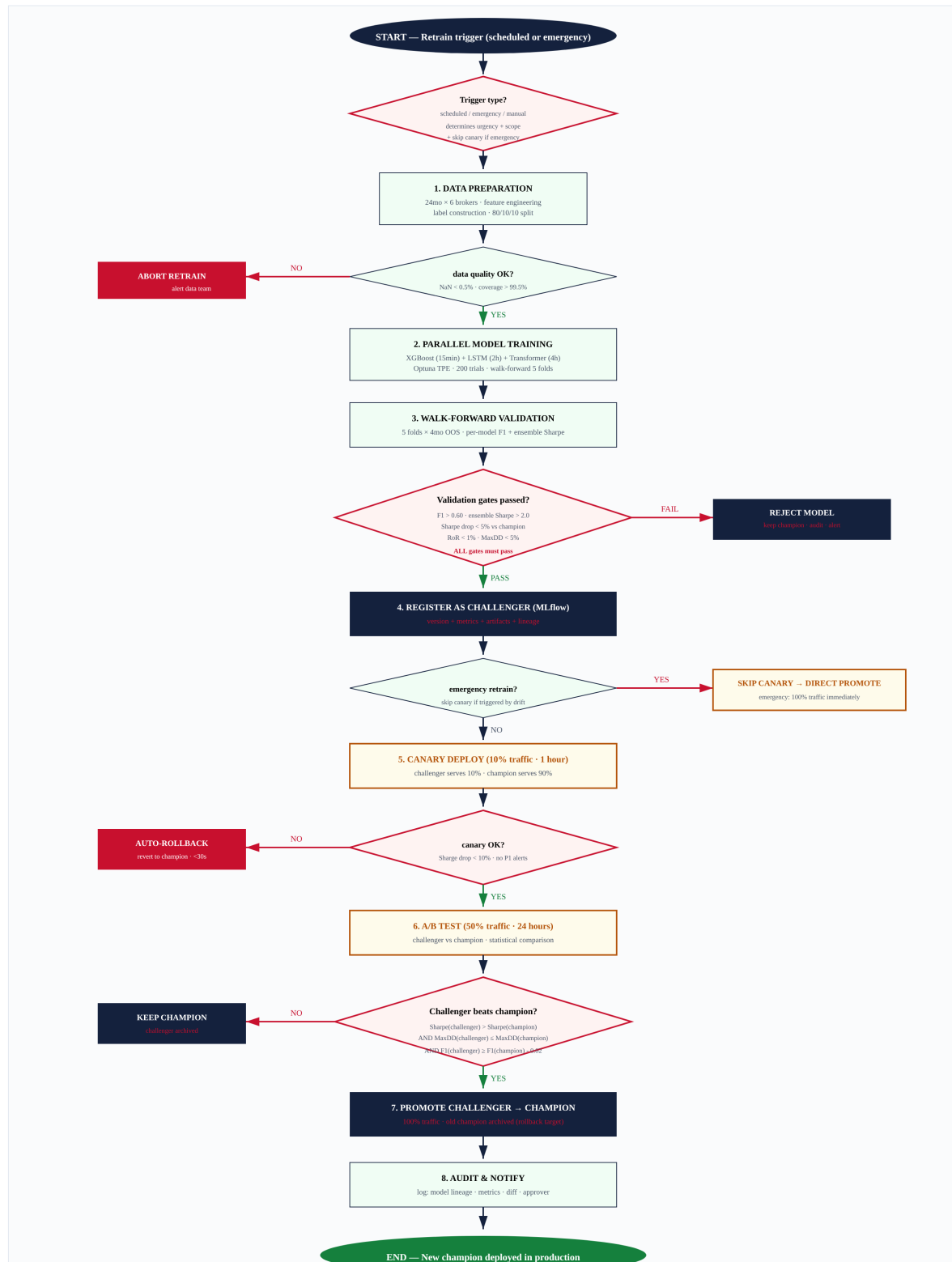


Figure 3.1 — End-to-end retraining workflow: trigger → data → train → validate → champion-challenger → deploy.

Weekly Retraining Schedule

Weekly Retrain (Sunday 22:00 UTC):

1. Data Preparation (30 min)
 - 24mo × 6 brokers historical data
 - Feature engineering (8 features)
 - Label construction (direction, trend_continue, regime context)
 - 80/10/10 train/val/test split
2. Parallel Model Training (~5h total)
 - XGBoost: 500 trees, Optuna TPE, 200 trials (~15 min)
 - LSTM: 2 layers, 128 hidden, Adam, 100 epochs (~2h)
 - Transformer: 4 heads, 2 layers, AdamW, warmup (~4h)
 - All 3 train in parallel on separate GPUs
3. Walk-Forward Validation (~30 min)
 - 5 folds × 4mo OOS
 - Per-model F1 > 0.60
 - Ensemble Sharpe > 2.0
 - Challenger Sharpe ≥ Champion - 5%
4. Register + Deploy (~30 min)
 - MLflow version + artifacts + lineage
 - Canary 10% (1h) → A/B 50% (24h) → Promote 100%
 - OR: Reject + keep champion + audit

Total: ~6 hours (Sunday 22:00 → Monday 04:00 UTC)

Monthly Retraining Schedule

Monthly retraining (1st Sunday) adds RL PPO retraining (8 hours, 10k episodes) and HMM per-session retraining (30 min per session × 3 sessions = 1.5 hours). The RL retrain uses a simulated XAUUSD environment with the latest market data, and the HMM retrain uses the last 90 days of session-specific data. Total monthly compute: ~12 hours (weekly 6h + additional RL + HMM 6h). The monthly retrain follows the same champion-challenger pipeline as weekly, but with separate validation gates for RL (episode reward > no-RL baseline) and HMM (regime F1 > 0.70).

Chapter 4 — Model Drift & Performance Decay Detection

Two independent detection systems run continuously: ModelDriftDetector (feature drift via PSI + concept drift via F1 drop) and PerformanceDecayDetector (Sharpe + DD + PF decay). Either system can trigger an emergency retrain independently.

03 Model Drift & Performance Decay Detection

PSI · concept drift · Sharpe decay · per-model monitoring



the no-RL baseline triggers an RL-only retrain.

Chapter 5 — Champion-Challenger & Version Control

The Champion-Challenger system ensures that no model is promoted to production without proving itself against the current champion. The challenger goes through a 3-stage pipeline: canary (10% traffic for 1 hour), A/B test (50% traffic for 24 hours), and promote (if it beats the champion). If it fails at any stage, it is archived and the champion is retained. Emergency retrains skip canary and A/B, deploying directly.

04 Champion-Challenger & Version Control

A/B testing · canary deploy · rollback < 30s · MLflow registry

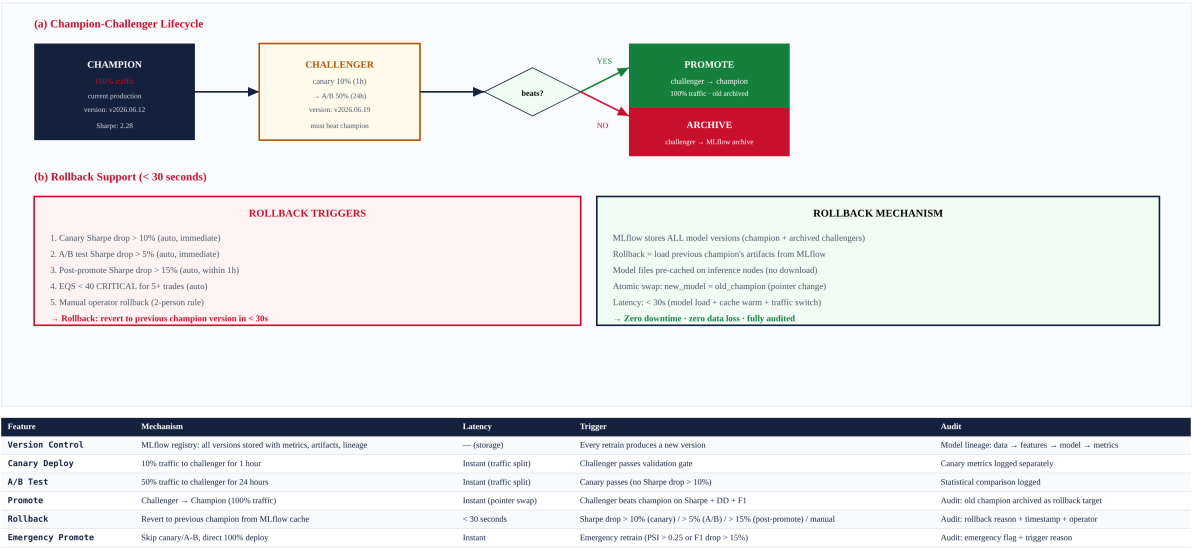


Figure 5.1 — Champion-challenger lifecycle, rollback triggers, and rollback mechanism.

Rollback Support

Rollback is always available. MLflow stores all model versions (champion + archived challengers). The previous champion's artifacts are pre-cached on inference nodes, so rollback is simply a pointer change — no model download needed. Rollback latency: < 30 seconds (model load + cache warm + traffic switch). Zero downtime, zero data loss, fully audited. Rollback triggers: canary Sharpe drop > 10%, A/B Sharpe drop > 5%, post-promote Sharpe drop > 15%, EQS < 40 CRITICAL for 5+ trades, or manual operator rollback (2-person rule).

Version Control (MLflow)

MLflow registry stores every model version with: version number, training data version, feature version, model artifacts (weights + config), training metrics (loss, F1, Sharpe), validation metrics (walk-forward results), and deployment decision (promoted / archived / rolled back). Full model lineage is maintained: data → features → model → metrics → decision. This enables full reproducibility — any past model can be recreated from its lineage chain.

Chapter 6 — Governance Framework

The RTF is governed by a board (CTO + Lead Quant + Risk Officer + ML Engineer) with a defined approval matrix. Most actions are automated (weekly retrain, canary, A/B, promote, auto-rollback), but certain actions require human approval (monthly retrain review, manual rollback, risk mode changes). All actions are audited in a hash-chained immutable log.

05 Governance & Monitoring Framework

Approval matrix • audit trail • 12-panel dashboard • alert routing

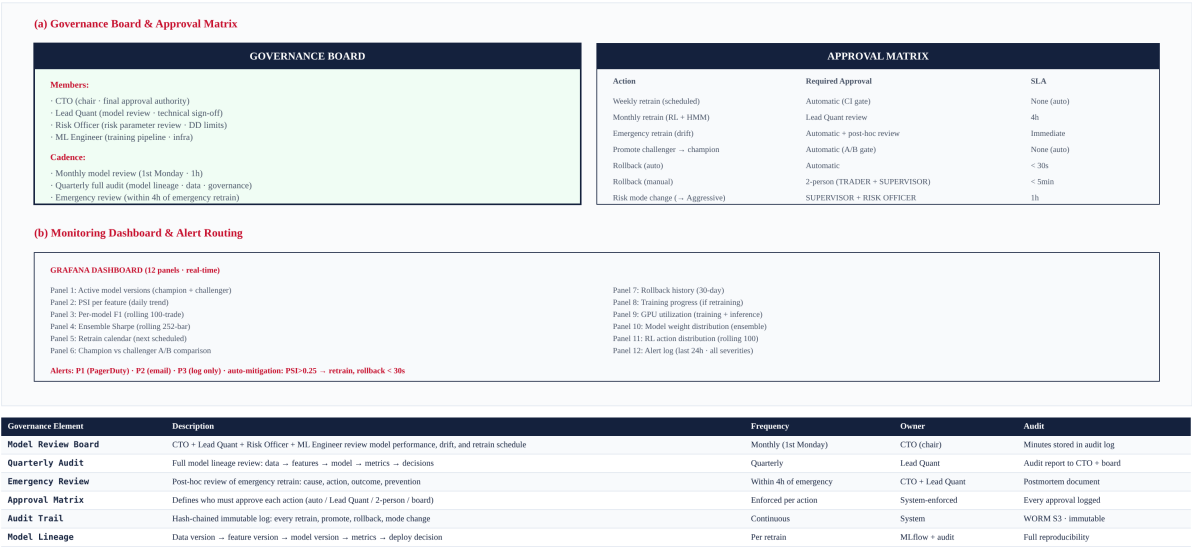


Figure 6.1 — Governance board, approval matrix, 12-panel dashboard, and alert routing.

Approval Matrix

- Weekly retrain: Automatic (CI gate only, no human approval needed)
- Monthly retrain: Lead Quant review (4h SLA)
- Emergency retrain: Automatic + post-hoc review (within 4h)
- Promote challenger: Automatic (A/B gate, no human approval)
- Auto-rollback: Automatic (< 30s, no human approval)
- Manual rollback: 2-person rule (TRADER + SUPERVISOR, < 5min)
- Risk mode change (to Aggressive/Competition): SUPERVISOR + RISK OFFICER (1h SLA)

Audit Trail

Every retrain, promote, rollback, and mode change is recorded in a hash-chained immutable audit log (WORM S3). Each entry includes: timestamp, action type, model version, trigger reason, metrics (before/after), approver identity, and model lineage. The audit log supports full regulatory compliance (MiFID-II trade reconstruction, model risk management) and post-incident forensic analysis.

Chapter 7 — Monitoring Framework

The RTF monitoring framework provides real-time visibility into all aspects of the model lifecycle. A 12-panel Grafana dashboard displays: active model versions, PSI per feature, per-model F1, ensemble Sharpe, retrain calendar, champion vs challenger A/B comparison, rollback history, training progress, GPU utilization, model weight distribution, RL action distribution, and alert log. Alerts are routed by severity: P1 (PagerDuty), P2 (email), P3 (log only).

Auto-Mitigation

The monitoring framework includes automatic mitigation: PSI > 0.25 triggers auto-retrain (skip canary); canary Sharpe drop > 10% triggers auto-rollback (< 30s); F1 drop > 15% triggers emergency retrain; performance degradation triggers weight reduction to floor (0.10). These auto-actions ensure the system adapts without waiting for human intervention.

Chapter 8 — Validation Process

Every retrained model must pass 8 CI gates before it can be promoted: (G1) per-model F1 > 0.60, (G2) ensemble Sharpe > 2.0, (G3) challenger Sharve >= champion - 5%, (G4) MaxDD < 5% and RoR < 1%, (G5) PSI < 0.20 (no drift after retrain), (G6) RL reward > no-RL baseline, (G7) rollback verified (< 30s), (G8) audit log complete. All 8 gates must pass — any failure rejects the model and retains the champion.

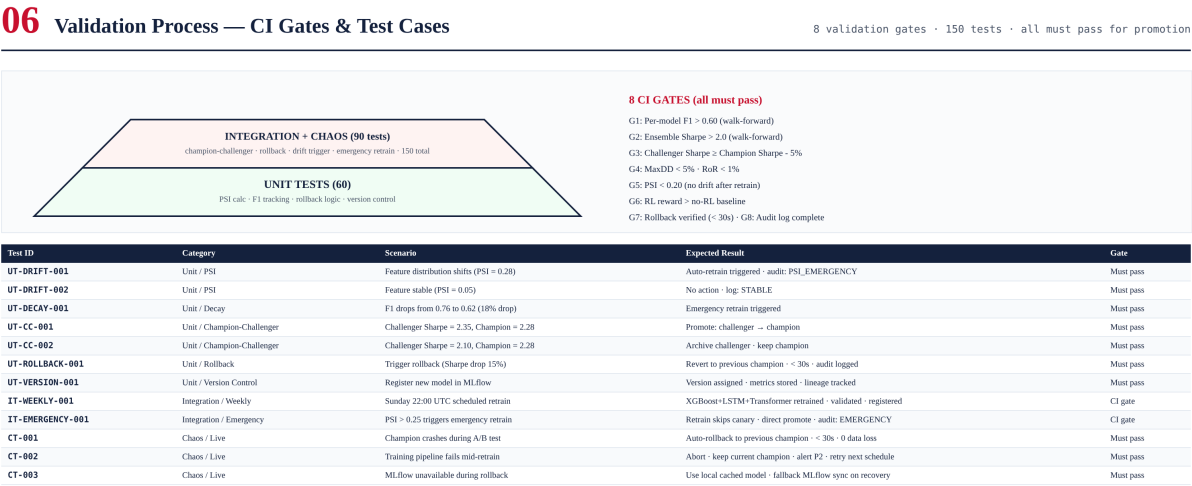


Figure 8.1 — Test pyramid (150 tests) and 8 CI gates.

Chapter 9 — Integration with TITAN Core

The RTF integrates with the Hybrid AI Stack (Module 7) as the model lifecycle manager. It consumes performance metrics from the AI Stack's monitoring framework, triggers retrains based on drift/decay signals,

and deploys new models via the champion-challenger pipeline. The RTF also integrates with the Risk Engine (Module 8) — if a hard DD breaker fires, the RTF reviews whether a model retrain is needed. The Execution Cost Intelligence (Module 9) feeds cost data to the RTF for cost-aware model evaluation (a model that produces more signals but at higher cost may not be promoted even if its Sharpe is higher).

Retrain Calendar Summary

Cadence	Models	Trigger	Duration	Canary?	Approval
Weekly (Sun 22:00 UTC)	XGBoost + LSTM + Transformer	Scheduled	~6 hours	Yes (10% → 50% → 100%)	Automatic (CI gate)
Monthly (1st Sun)	RL PPO + HMM (all sessions)	Scheduled	~12 hours (incl weekly)	Yes (separate track)	Lead Quant review
Emergency (anytime)	Drifted model(s) only	PSI > 0.25 or F1 drop > 15%	~2-6 hours	No (direct promote)	Automatic + post-hoc
Manual (anytime)	Specified by operator	Operator request	~2-6 hours	Yes	2-person (TRADER + SUPERVISOR)